W06

Spatial data in R with sf and terra

R PROGRAMMING · GIS & MAPPING

You repeat in code what you did in QGIS last week. With `sf` (vectors) and `terra` (rasters) the analysis is written step by step: it can be checked, shared and rerun on new data.

DURATION	About 2 hours.
PACKAGES	`sf`, `terra`, `tidyverse`
FILES	`week06_sf_and_terra.R`, `week06_solutions.R`, `montfred.gpkg`, `dem.tif`, `landcover.tif`, `landcover_classes.csv`, `survey_points.csv`
DATA	The same as Week 5.

Learning outcomes

- read GeoPackages and GeoTIFFs and check their CRS;
- turn a CSV with coordinates into points and reproject them;
- run vector operations: areas, intersections, buffers, distances;
- crop, mask, reclassify and derive terrain from a raster;
- extract raster values at points and in zones around them.

1 · From QGIS to R

Task	QGIS	R
Read vector layer	Add Vector Layer	`st_read()`
Read raster	Add Raster Layer	`rast()`
CSV to points	Delimited Text Layer	`st_as_sf(coords = ...)`
Reproject	Save As, other CRS	`st_transform(), project()`
Buffer	Buffer	`st_buffer()`
Select by location	Select by Location	`st_intersects()`
Clip raster	Clip Raster by Mask	`crop() + mask()`
Slope, hillshade	Raster > Analysis	`terrain(), shade()`
Reclassify	Reclassify by table	`classify()`
Values at points	Sample raster values	`extract()`

2 · Vectors with sf

An sf object is a data frame with a special geometry column: everything you know from dplyr works on it.

```
park <- st_read("data/montfred.gpkg", layer = "park_boundary")
pts  <- read_csv("data/survey_points.csv") |>
  st_as_sf(coords = c("longitude", "latitude"), crs = 4326) |>   # degrees
  st_transform(st_crs(park))                                       # to UTM, metres
as.numeric(st_area(park)) / 10000                                  # hectares
pts$in_park <- lengths(st_intersects(pts, park)) > 0
riparian <- st_buffer(streams, 150) |> st_union()
```

Common mistake

If you buffer or measure distances with points in EPSG:4326, the units are degrees. sf warns in some cases but not all. Always check `st_crs()` before measuring.

3 · Rasters with terra

```
dem  <- rast("data/dem.tif")
slope <- terrain(dem, "slope", unit = "degrees")
dem_park <- crop(dem, park) |> mask(vect(park))   # vect() converts sf to terra
rcl <- matrix(c(-Inf, 600, 1,  600, 1000, 2,  1000, 1300, 3,  1300, Inf, 4),
              ncol = 3, byrow = TRUE)
belts <- classify(dem_park, rcl)                  # elevation belts
```

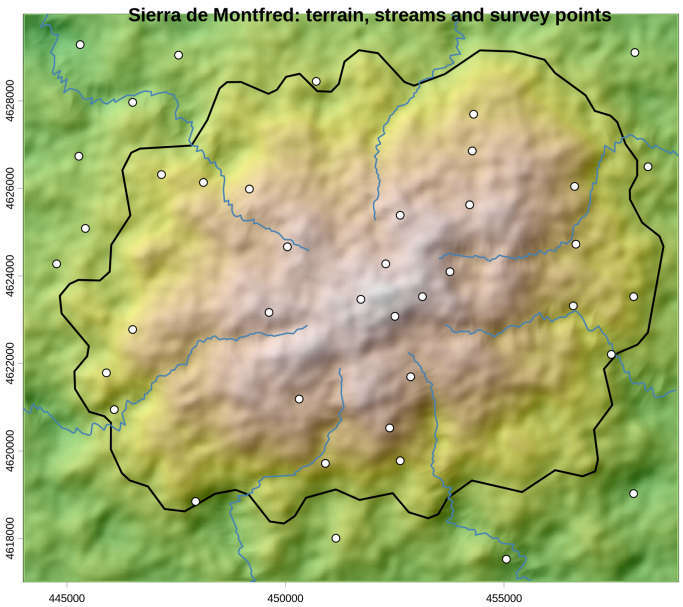
4 · Extracting values

`extract()` returns the cell value under each point. With a polygon (for example a 250 m buffer around each point) and `fun = mean`, it returns the zone's mean: this gives landscape variables such as the proportion of open habitat around each point, which we use in models.

```
pts$dem_m    <- terra::extract(dem, vect(pts))[, 2]
open         <- lc %in% c(5, 6)                       # heathland or meadow
pts$open_250 <- terra::extract(open, vect(st_buffer(pts, 250)), fun = mean)[, 2]
```

terra::extract and tidyr::extract

The tidyverse also has an `extract()` function. To avoid confusion, write `terra::extract()`.



Hillshade, semi-transparent DEM, park, streams and points, drawn with terra and sf plot().

Exercises

Answers in `week06_solutions.R`.

1. How many points are within 500 m of the park boundary? (`st_boundary()`, `st_distance()`)
2. Area (ha) of each elevation belt.
3. Mean slope of the park versus the riparian zone.
4. Proportion of beech forest within 500 m of each point. Which point has the most?
5. Save the elevation belts as a GeoTIFF and open them in QGIS. Do they match Week 5?

Before Week 7

- Install `install.packages("tmap")` (version 4).
- If you have a university ArcGIS Pro licence, sign in once before class.